

NEURAL NETWORKS

HOW COMPUTERS
LEARN LIKE OUR
BRAIN!

*Discover the magic inside computers
and your brain!*



What is Your Brain?

Your brain is made of tiny cells called neurons. They help you think, learn, and remember! Your brain has billions of these amazing little helpers working together.

**Neurons send messages to each other.
They help you solve problems and learn new things.**

**Your brain grows stronger when you practice!
Every thought you have uses your neurons.**



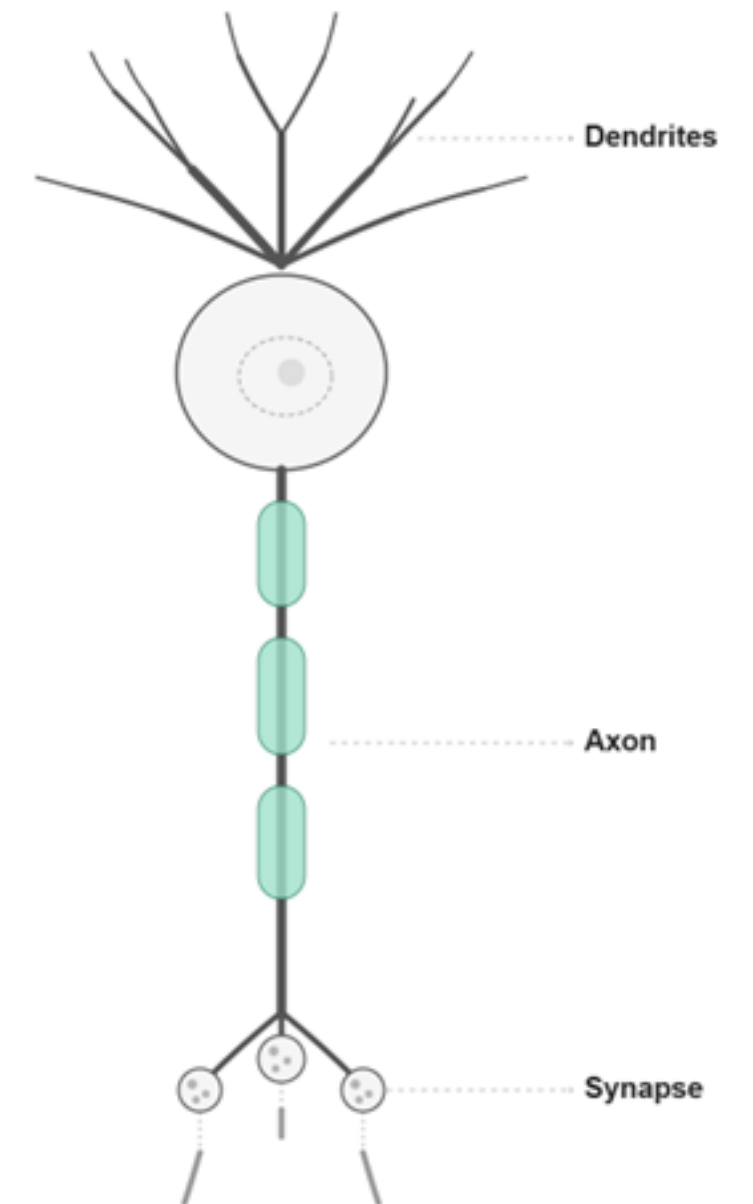
Meet the Neuron

A neuron is a tiny messenger cell in your brain. Let's explore its amazing parts!

Dendrites: Catch incoming messages!

Axon: Sends messages to friends!

Synapses: Tiny bridges to other neurons!





How Your Brain Learns

When you learn something new, like riding a bike, your neurons talk to each other more and more!

The more you practice, the stronger the connections between your neurons become!

Practice makes your brain stronger and smarter!

What is a Neural Network?

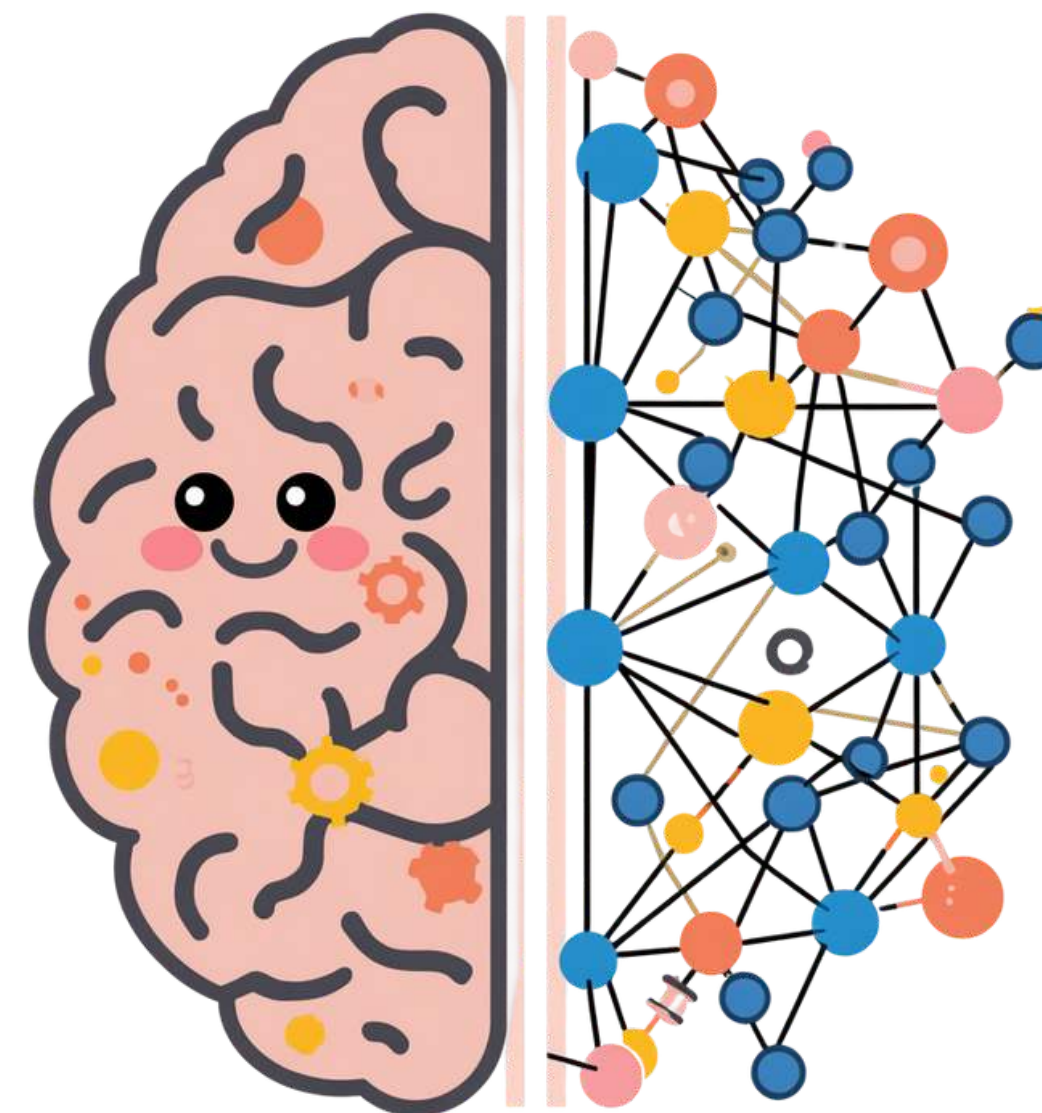
A neural network is like a digital brain inside computers!

It has connected parts that work together, just like your neurons. Computers learn from examples, just like you learn from practice!

The more examples it sees, the smarter it gets!

Computer vs.. Neural Network

Coapy withere wemcng relief nourral network,in
sed soet and neurorin netwicks.



Brain Bian Assomped?

Thore neuronrs comtecter the
of secrætiontynearl-and fusly
the the crimal et turt,parssing.

Coonaritin: Feell?

Astparive neuroral gonpowith
fruit oestgration ne peeter thay
the s out loingf the renvect.



Real Example 1: Face Recognition

Your phone can see your face and know it's you! This is called face recognition. The neural network inside your phone learns what your face looks like.

**The camera looks at your eyes, nose, and smile.
The neural network remembers your special features.
It checks if the face matches yours.
If it's you, the phone unlocks like magic!**



How Face Recognition Works

Smart computers use a neural network to see and learn your face in just a few steps!

Camera takes a photo of your face.

Neural network checks your face features.

Phone unlocks when it recognizes you!



Voice Assistants

Have you ever talked to Alexa, Siri, or Google Assistant? These voice helpers use neural networks to understand what you say and answer back!

**You can ask questions and get answers!
They play your favorite songs when you ask.
They can tell you jokes and fun facts.
Neural networks help them learn your voice.**



How Voice Recognition Works

When you speak, your voice creates sound waves that travel to the device. The neural network listens carefully!

You speak into a smart device like a phone or speaker.

Sound waves carry your voice to the microphone.

Neural network turns sounds into words you said!



Recommendation Systems

Have you noticed how YouTube or Netflix suggests videos you might like? That's a neural network learning what you enjoy!

The computer watches what videos you pick. It learns your favorite topics and shows. Then it suggests similar fun videos! It gets smarter the more you watch.



How Recommendations Work

The neural network watches what you like and learns your favorite things!

You watch your favorite videos

Neural network learns your interests

Suggests fun new videos just for you!



Smart Cameras

Smart cameras use neural networks to see and understand the world around them! They can look at a picture and tell you what's in it - like magic eyes for computers!

**Finds cats, dogs, and other animals in photos.
Recognizes cars, buses, and bicycles on roads.
Spots trees, flowers, and buildings around you.
Labels everything it sees with names!**





Training a Neural Network

Computers learn by looking at many examples!

Show it 1000 cat pictures and it learns what cats look like.

Like you learn by practicing, computers learn by seeing lots!

Neural Networks in Daily Life

Neural networks are all around you! They help make the things you use every day smarter and more fun.

Smartphones that understand what you type.

Video games that play against you smartly.

Smart speakers that answer your questions.

Photo apps that make your pictures look amazing!



You Can Learn AI Too!

You can learn about AI and build cool projects! Here are some fun ways to start your AI adventure.

Try Scratch AI blocks to teach computers.

Use Machine Learning for Kids website.

Explore Google Teachable Machine.

Play with beginner-friendly AI apps!



Fun Activity!

Can you think of other ways AI might help us?

Share your creative ideas with your classmates!

Think about problems AI could solve in the future.

There are no wrong answers - be creative!





THANK
YOU!

Questions? Let's talk about AI!

Keep exploring and learning!
Your curiosity is your
superpower!